

Shihan Kanungo

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San José State University

Concurrent enrollment during high school. Graduate courses indicated by †

A+	MATH 279A † Graph Theory. Yan Zhang	Spring 2024
	Douglas B. West. <i>Introduction to Graph Theory</i> . Prentice Hall 1996	
A	MATH 221A † Higher Algebra I. Edgar Bering	Fall 2025
	Paulo Aluffi. <i>Algebra: Chapter 0</i> . AMS 2021. Covered Chs. 1–4.	
A+	MATH 221B † Higher Algebra II. Edgar Bering	Spring 2026
	Paulo Aluffi. <i>Algebra: Chapter 0</i> . AMS 2021. Covered Chs. 5–9.	
A+	MATH 226 † Theory of Numbers. Jordan Schettler	Spring 2024
	Primes in arithmetic progressions, partitions, modular group, and the Dedekind eta function.	
A+	MATH 285M † Elliptic Curves and Modular Forms. Jordan Schettler	Fall 2025
	Alvaro Lozano-Robledo. <i>Elliptic Curves, Modular Forms, and Their L-functions</i> . AMS 2011.	
A+	MATH 229 † Advanced Matrix Theory. Wasin So	Fall 2024
	R. Horn & C. Johnson. <i>Matrix Analysis</i> , Cambridge University Press 2012.	
A+	MATH 285M † Nonnegative matrices. Wasin So	Spring 2025
	The basic theory of nonnegative matrices; applications to graphs and digraphs.	
A+	MATH 275A † Topology. Yan Zhang	Fall 2025
	Compactness, connectedness; Tychonoff's Theorem; paracompactness and metrization theorems.	
A+	MATH 231A † Real Analysis. Slobodan Simic	Fall 2024
	E.M.Stein & R.Shakarchi, <i>Real Analysis: Measure Theory, Integration, and Hilbert Spaces</i> , Chs. 1–4.	
A+	MATH 231B † Functional Analysis. Slobodan Simic	Spring 2025
	L Debnath & P Mikusinski. <i>Introduction to Hilbert Spaces with Applications</i> . Academic Press 2005.	
A+	MATH 238 † Advanced Complex Variables. Jordan Schettler	Spring 2026
	E.M.Stein & R.Shakarchi, <i>Complex Analysis</i> , Chs. 1–5, 8.	
A+	MATH 233A † Partial Differential Equations. Daniel Brinkman	Spring 2025
	Peter J Olver. <i>Introduction to Partial Differential Equations</i> . Springer 2013.	
A+	MATH 234 † Advanced Dynamical Systems. Liam Stanton	Spring 2024
	Continuous and discrete systems; stability of equilibria and closed orbits, structural stability.	
A	MATH 285A † Stochastic Methods in Scientific Computing. Liam Stanton	Spring 2025
	Monte Carlo methods, Metropolis-Hastings algorithm, random-number generation.	
A	MATH 129A&B Linear Algebra I & II. Dashiell Fryer/Wasin So	2023-24
	S.H.Friedberg, et al. <i>Linear Algebra</i> 2nd ed., Prentice Hall 1989.	
A+	MATH 131A&B Introduction to Analysis I & II. Tim Hsu	2024
	Kenneth Ross, <i>Elementary Analysis: The Theory of Calculus</i> . UTM, Springer NY 2013.	
	Tim Hsu, <i>Fourier Series, Fourier Transforms, and Function Spaces</i> . AMS/MAA 2020	
A+	MATH 138 Complex Variables. Liam Stanton	Fall 2023
	Dennis G. Zill et al, <i>A First Course in Complex Analysis with Applications</i> . Jones & Bartlett 2006.	
A+	MATH 179 Introduction to Graph Theory. Sogol Jahanbekan	Fall 2024
	R. J. Wilson. <i>Introduction to Graph Theory</i> . Prentice Hall 2010	

Euler Circle

College-level math classes taught by Simon Rubinstein-Salzedo. Grades not assigned.

Abstract Algebra.

Spring 2020

Group theory, Sylow theorems, fields and extensions, number fields, Galois correspondence.

Combinatorics.

Fall 2025

Special numbers; sieving; restricted permutations; posets; incidence algebras; generating functions.

Combinatorial Game Theory.

Fall 2024

Nim, Hackenbush, surreal numbers, impartial games, Sprague-Grundy theory.

Generating Functions.

Fall 2021

Philippe Flajolet and Robert Sedgewick. *Analytic Combinatorics*. Cambridge University Press 2011.

Number Theory.

Fall 2023

Reciprocity theorems, quadratic forms, elliptic curves, modular curves.

Analytic Number Theory.

Spring 2024

Dirichlet's Theorem, The Prime Number Theorem, Brun's theorem, smooth numbers.

Ergodic Theory.

Winter 2024

Simon Rubinstein-Salzedo. *Ergodic Theory*, The Carus Mathematical Monographs. MAA Press 2025.

Differential Geometry.

Spring 2022

Andrew Pressley. *Elementary Differential Geometry*. Springer, 2001.

Markov Chains.

Fall 2020

D.A. Levin et al, *Markov chains and mixing times*. American Mathematical Society 2006

Representation Theory

I haven't taken a formal course, but have studied the subject through the following books.

Introduction to Lie algebras.

2006

Karin Erdmann. Springer Verlag, London.

Introduction to Lie Algebras and Representation Theory.

1972

Humphreys, J.E. Springer Verlag, New York.

Representations of Semisimple Lie Algebras in the BGG Category $\mathcal{O}$.

2008

Humphreys, J.E. Graduate Studies in Mathematics **94**, AMS.

Lie Superalgebras and Enveloping Algebras.

2012

Ian. M. Musson. Graduate Studies in Mathematics **131**, AMS.

Yangians and Classical Lie algebras.

2007

Alexander Molev. Mathematical Surveys and Monographs **143**, AMS.

Dualities and Representations of Lie Superalgebras.

2012

Cheng, S. and Wang, W. Graduate Studies in Mathematics **144**, AMS.

Lectures on Symmetric Tensor Categories.

2021

Pavel Etingof and Arun Kannan. [[arXiv](#)]